

AirTwinNet – System Architecture & Project Summary

A Digital Twin Enabled Multi-Hybrid Framework Combining Satellite Imagery, IoT Sensors, Graph Neural Networks, and Explainable AI for Urban Air Quality Prediction

1. Project Overview

AirTwinNet is a multi-modal, spatio-temporal framework for urban air-quality prediction. It combines satellite observations, IoT sensor measurements, meteorological information, historical air-quality data, an Urban Air Quality Digital Twin, Spatio-Temporal Graph Neural Networks (ST-GNN), hybrid prediction, and Explainable AI (XAI).

2. Core Architecture Flow

- Physical Urban Environment → Satellite, IoT, weather, and historical air-quality data
- Data ingestion & preprocessing → Spatio-temporal alignment
- Multi-modal data fusion → Unified spatio-temporal representation
- Urban Air Quality Digital Twin → Dynamic representation of the physical city state
- Dynamic urban air-quality graph → Nodes, edges, features, and temporal relationships
- ST-GNN → Spatial and temporal representation learning
- Multi-hybrid prediction engine → PM2.5, PM10, AQI and other pollutant forecasts
- Prediction confidence / uncertainty → Reliability indication
- Explainable AI → Feature, spatial, temporal, sensor and satellite contribution analysis
- Decision-support dashboard → Maps, hotspots, forecasts, alerts and explanations
- Feedback & retraining → Actual-vs-predicted comparison, error monitoring and model updates

3. Role of the Digital Twin

The Urban Air Quality Digital Twin is the central innovation of AirTwinNet. It provides a virtual representation of the urban environment and maintains real-time and historical state, sensor locations, pollution distribution, and environmental conditions. IoT and environmental observations synchronize the twin, while the twin supports prediction, simulation, and what-if analysis.

4. Graph Learning & Prediction

The fused urban data is represented as a dynamic graph. Nodes represent sensors, urban locations, or spatial cells; edges represent relationships such as spatial proximity, wind influence, pollution correlation, or relevant urban connectivity. Node features can include pollutant, satellite, and meteorological information. The ST-GNN learns spatial dependencies and temporal pollution patterns before the hybrid prediction engine produces forecasts.

5. Explainable AI & Decision Support

XAI explains why a prediction was produced using mechanisms such as SHAP, feature importance, attention analysis, and spatial/temporal contribution analysis. The dashboard can present AQI and pollutant forecasts, pollution hotspots, risk levels, alerts, confidence information, and interpretable insights.

6. Feedback & Continuous Improvement

The architecture closes the loop by comparing actual IoT observations with predictions. Prediction errors are monitored and can support model updating/retraining and synchronization of the Digital Twin, creating a continuous improvement cycle.

Final System Architecture

